



Human-Monkey hybrids

A team of scientists have begun mixing human cells into monkey embryos to create hybrids.
<https://digit.in/sep19-01>



Child cognitive testing

Research has indicated that cognitive testing at the age of 3 can reveal how their brain will age.
<https://digit.in/sep19-02>

INSULAR EVOLUTION

How islands influence evolution

Aditya Madanapalle | aditya@digit.in

W

hen animals cross water bodies and colonise islands, strange things start happening to them

over generations. Bigger animals tend to grow smaller, while smaller animals tend to grow larger. The phenomenon has been noticed in all kinds of living things, mammals, birds, reptiles, the extinct dinosaurs and even plants. The first attempt at exploring the phenomenon was made by JB Foster. Foster published tables showing some species, such as deer and rabbits grow smaller on islands. Other species, such as rodents, tended to grow larger. The data was interpreted to indicate that in island environments, small creatures tended to grow larger, while larger creatures tended to grow smaller. This is known as Foster's Rule as well as the Island Effect.

In the absence of predators, animals also start losing their ability to respond to threats. Here is how islands influence evolution.

DWARFISM

There are many theories on what causes insular dwarfism, which is the tendency of animals to grow smaller on islands. A lack of predators may remove the pressures that caused animals to grow large in the first place, as a countermeasure against being easily caught and eaten. The digestion in smaller herbivores is more efficient. Additionally, there may be limited resources on an island for a

groups of large herbivores, such as elephants. The limited area available on islands means that groups or individuals can only occupy smaller territories. In the lower latitudes, thermoregulation is easy for smaller animals. In mammals, the general trend is that the smaller the size, the more offspring are produced over an individual's life span. All of these factors may offer advantages in island environments, and are not mutually exclusive.

The most dramatic examples of island dwarfism are in creatures that are known to be large herbivores. The magyarsaurus, was a sauropod like the Brontosaurus or the Apatosaurus, but was only as tall



A pygmy leaf chameleon from Madagascar

Image: Charles J Sharp